Failure-Trace Allocation for Budgeted Large Language Model Inference

Soroush Vahidi

Department of Computer Science, New Jersey Institute of Technology, Newark, NJ, USA
sv96@njit.edu
ORCID: 0000-0003-1934-6282

Abstract

We study *matched-budget answer selection*: given candidate answers already generated by large language models under an equal per-method inference budget, the task is to surface the best answer without additional model calls and without access to ground-truth labels at decision time. We present **Failure-Trace Allocator (FTA)**, a deterministic two-gate post-generation policy that exploits failure-trace metadata logged as a side-effect of the frontier generation run. The *Low-Depth Guard* fires when the frontier search terminates on a structurally weak single branch, substituting an external majority-vote answer. The *External-Consensus Fallback* fires, as a low-frequency high-precision safeguard, when all three independent external baselines unanimously disagree with the frontier's selected answer. Both gates are strictly gold-free and issue no additional inference calls.

We evaluate on the GSM8K benchmark [1] using Cohere command-r-plus-08-2024 under a matched per-method cap of six logical inference calls per example. On the primary 300-example evaluation set, FTA achieves **86.67%** accuracy. On a disjoint 720-example aggregate spanning three independent evaluation seeds, FTA achieves **80.69%** accuracy versus 77.64%, 77.08%, and 75.14% for three independently sourced external baselines (L1, S1, and TALE). Source-stratified 95% bootstrap confidence intervals are strictly positive for all four pairwise comparisons, with the lower bound versus the best external at +1.11 percentage points. As supporting evidence, a multi-provider regime-learning selector trained on the same failure-trace signal improves over the frontier under matched conditions with zero false-override events. Against a pooled four-answer ensemble, FTA remains ahead by point estimate but is not statistically separated on either evaluation set. All findings are scoped to the tested provider and benchmark; generalization to other providers or tasks requires independent replication.

Keywords: Large language models budgeted inference answer selection failure-trace metadata test-time compute

1 Introduction

Scaling inference-time computation has emerged as a practical strategy for improving the accuracy of large language models on reasoning tasks without additional model training [2]. By generating multiple candidate answers under a fixed inference budget and selecting among them, a system can achieve accuracy gains comparable to those from model-parameter scaling, at inference time [3]. This reframes answer quality as an *intelligent decision problem*: given a pool of already-generated candidates, which selection rule best exploits the available runtime information under cost constraints?

In settings where the inference budget is *matched per method*—every candidate-producing method receives the same number of model calls per example—the quality of the final answer depends entirely on the selection rule applied to the already-generated pool. Simple majority vote treats all candidates equally; yet a frontier generation run produces structured metadata—records of search depth, branch structure, and override events—that is richer than the raw candidates alone and may carry reliable signal about candidate trustworthiness. Exploiting such runtime-visible signals for dynamic selection is the core question studied here.

We examine this question in a concrete, reproducible setting. The task is grade-school arithmetic reasoning on the GSM8K benchmark [1], the provider is Cohere command-r-plus-08-2024, and the inference budget is $B = 6$ logical calls per example. A frontier generator produces candidate answers via program-aided execution [4]; three independent external methods (L1, S1, and TALE) provide reference answers under the same budget cap. Given these already-generated outputs, the goal is to select a final answer without additional inference calls and without using gold labels.

Contributions. The main contribution is **Failure-Trace Allocator (FTA)**, a deterministic two-gate post-generation policy that uses failure-trace metadata logged during normal frontier execution:

- The *Low-Depth Guard* fires when the frontier's execution trace signals a structurally weak single-branch output, substituting an external majority-vote answer.
- The *External-Consensus Fallback* fires when all three external baselines unanimously disagree with the frontier's selected answer.

Both gates are strictly gold-free: no correctness labels, ground-truth answers, or example identifiers are accessed at decision time. The two gates operate in strict priority order; at most one fires per example.

On the primary evaluation set of 300 disjoint examples (Final-300), FTA achieves **86.67%** accuracy. On a disjoint Aggregate-720 combining three independent evaluation seeds (seeds 41, 61, and 71; $300 + 120 + 300$ examples with zero overlap in example identifiers or question hashes), FTA achieves **80.69%** accuracy versus 77.64% (L1), 77.08% (S1), and 75.14% (TALE). Source-stratified 95% bootstrap confidence intervals are strictly positive for all four pairwise comparisons, including versus the best external source (CI lower bound +1.11 pp). As supporting evidence, a multi-provider regime-learning selector (D9) trained on the same failure-trace signal improves over the frontier under matched conditions with zero false-override events. FTA remains ahead of a pooled four-answer ensemble (frontier+L1+S1+TALE) by point estimate on both evaluation sets, but that comparison is not statistically separated.

We also document five not-retained prototype variants evaluated against the same criteria; these are reported as negative or insufficient results in Section 5.1 and Table 10.

Scope disclosures. Four scope notes are integral to interpreting all claims in this paper. *(i)* Confidence intervals are source-stratified bootstrap; the interval versus the pooled four-answer ensemble includes zero on both Final-300 and Aggregate-720, so superiority over ensembling is *not* claimed. *(ii)* The matched-budget contract is an evaluation design: constructing the full frontier+L1+S1+TALE pool from scratch corresponds to 24 logical calls per example, not six. *(iii)* All evaluation and selection findings are scoped to GSM8K with the Cohere command-r-plus-08-2024 provider; generalization to other benchmarks or providers requires independent replication. *(iv)* The 120-example evaluation set at seed 61 is failure-enriched by construction; results on seeds 41 and 71 serve as out-of-distribution checks.

Paper organization. Section 2 reviews related work on test-time compute scaling, verifier-guided reranking, selective prediction, and cost-aware routing. Section 3.3 specifies FTA precisely, including exact trigger conditions and a policy summary table. Sections 4 and 5 describe the evaluation protocol and report results with full confidence intervals. Section 5.1 summarizes ablation variants and not-retained outcomes. Sections 6–7 discuss implications, limitations, and directions for future work.

2 Related Work

Reasoning-time sampling, search, and self-consistency. Chain-of-thought prompting [5] and zero-shot chain-of-thought [6] showed that explicitly eliciting intermediate reasoning improves large-language-model performance on mathematical and symbolic tasks. Self-consistency extends this idea by sampling multiple reasoning paths and returning the most frequent final answer [3]. Follow-up work studies broader consensus variants, including universal self-consistency across generated candidates [7]. Tree-structured reasoning methods, including Tree of Thoughts [8] and Graph of Thoughts [9], further broaden inference-time exploration by organizing candidate reasoning paths into explicit search structures. Program-aided language-model reasoning similarly generates executable or semi-executable solution traces before extracting a final answer [4]. The present work is closest to this family in that it assumes multiple candidate answers and runtime traces are available, but it studies a different question: how to choose among already generated answers using gold-free execution metadata rather than selecting by raw frequency alone.

Budgeted inference and test-time compute. Recent work on test-time scaling studies how additional inference computation can improve reasoning accuracy [2]. Budget-forcing methods such as s1 shape the amount of reasoning performed during generation by extending or truncating traces toward a target budget [10], while broader analyses of inference-time interventions compare sampling, voting, search, and allocation strategies under cost constraints [11]. Our experiments include prompt-budgeting and support-based answer-selection baselines as evaluated comparators; they are introduced in the experimental setup rather than treated as established prior work. The contribution here is therefore not a new generation-time scaling law; it is a post-generation answer-selection rule evaluated after candidate-producing methods have run under the same per-method budget cap.

Verifier-guided reranking and answer selection. A complementary line of work trains verifiers or reward models to rerank candidate solutions. For GSM8K, Cobbe et al. [1] showed that an outcome verifier can substantially improve answer selection when trained on labeled solution data. Process reward models provide finer-grained supervision by scoring intermediate reasoning steps rather than only final answers [12]. Related lines include self-verification and iterative self-correction, where models critique or re-check their own outputs before finalizing an answer [13, 14]. Lightweight supervised text-classification frameworks such as SetFit can also support task-specific reranking when suitable labels are available [15]. In contrast, the policy studied here does not train a verifier and does not use correctness labels at decision time; it relies on runtime-visible frontier metadata and agreement patterns among independently generated candidate answers.

Ensembling, selective prediction, and calibration. Classical ensemble methods establish why simple voting can be robust when model errors are partially independent [16]. Selective prediction methods formalize abstention or fallback on subsets where a primary predictor is less reliable [17]. Calibration work studies when confidence-like quantities are trustworthy enough for such selective decisions [18]. FTA is closer to this selective-routing view than to pure ensembling: it keeps the frontier default on most examples and overrides only on a small runtime-identifiable subset.

Cost-aware routing, adaptive computation, and positioning. Cost-aware routing systems such as FrugalGPT [19] and language-model cascades [20] reduce inference cost by deciding which model or stage to invoke for a given input. Adaptive computation methods make a related decision within a model or computation graph, allocating more processing to inputs that appear to require it [21]. The setting in this paper is different: the evaluated candidate-producing methods are compared under a matched per-method budget, and Failure-Trace Allocator (FTA) itself is a deterministic selection layer that issues no additional model calls once candidate answers and metadata are available. The resulting claim is intentionally narrow: on the tested Cohere/GSM8K setting, two runtime-legal signals—weak frontier traces and unanimous external disagreement—provide useful information for selecting among already generated answers.

3 Problem Setup and Method

3.1 Problem Setup

We study *fixed-pool answer selection under a matched-budget inference contract.* The task is: given a set of candidate answers already produced by a collection of inference methods under an equal per-method budget, select the best final answer without issuing additional provider calls and without access to ground-truth labels at decision time.

Problem instance. Each instance comprises a question q, a *candidate pool* $\mathcal{C}(q) = \{a_F, a_{L1}, a_{S1}, a_T\}$ of four final-answer candidates generated under the matched-budget protocol (Section 4), and failure-trace metadata $m_F(q)$ produced as a byproduct of the frontier method's normal execution. A ground-truth answer $y(q)$ exists but is withheld at runtime; it is used only for offline evaluation. The selection policy maps $(\mathcal{C}(q), m_F(q)) \to \hat{a}(q) \in \mathcal{C}(q)$.

Candidate pool and budget accounting. The candidate pool is constructed once per evaluation run. The four candidate-generating methods each produce one final-answer candidate per example under a cap of $B = 6$ logical API calls: the frontier method and three external baselines (L1, S1, and TALE). Full candidate-pool construction therefore requires $4 \times B = 24$ logical inference calls per example. The answer-selection policy (FTA) is applied *after* pool construction: it selects among already-generated candidates and issues *no* additional inference calls.

Runtime-visible vs. offline-only information. All inputs to the selection policy are runtime-visible and gold-free: the frontier answer, the three external answers, and failure-trace metadata derived from the frontier method's logged execution. Ground-truth labels, exact-match signals, and correctness indicators are *never* inputs to any policy decision; they are used only offline for evaluation and failure analysis. The oracle accuracy (92.00% on Final-300) is a diagnostic upper bound computed offline with gold labels; it is not a comparison baseline.

Evaluation results are reported on:

- Final-300 (seed 71, $n = 300$): the primary unbiased evaluation set,
- Aggregate-720 (seeds 41, 61, 71; $n = 300 + 120 + 300$): the primary claim set combining three disjoint sources, with zero example-id and question-hash overlap across sources,
- Seed 61 ($n = 120$): a failure-enriched development split; absolute accuracy on this split is not representative of standard evaluation conditions and is not used to derive primary claims.

3.2 Terminology

The following terms are used consistently throughout the paper. Short paper-facing names are used for readability; the corresponding implementation identifiers are preserved in Table 16 for reproducibility.

Candidate pool
The set of final answers produced by the available methods (frontier and external baselines) for a given example. Candidate-pool construction is separate from the downstream selection policy.

Frontier method
The primary inference method whose answer serves as the default output before any policy gate fires. It produces failure-trace metadata as a side-effect of normal execution, including structural indicators of search breadth and internal answer agreement.

Failure-trace metadata
Runtime-visible fields logged during frontier execution that characterize the structural quality of the frontier's search output (e.g., whether the selected answer arises from a single weak branch or from an internally consistent exploration). These fields are gold-free by construction.

External baseline
Any of the three independently-sourced single-model methods used as reference signals: L1, S1, or TALE. External baselines provide gold-free consensus signals without access to gold labels.

L1 An external method that selects the answer with the highest support count across sampled model outputs.

S1 An external method that applies sequential budget-forcing to elicit a final answer within the allocated budget.

TALE An external method that uses prompt-level budget cues to guide answer selection.

Logical API call
A single provider inference request counted against a candidate-producing method's budget.

Budget The integer upper bound $B = 6$ on logical API calls per example for each candidate-producing method, applied identically to all compared methods (matched-budget condition).

Matched budget
All compared candidate-producing methods operate under the same per-method budget cap B, ensuring that accuracy differences are not confounded by unequal per-method computational limits.

Gold-free runtime feature
Any input to a policy decision rule derived solely from model outputs, run metadata, or structural properties of the candidate pool—never from gold labels, exact-match signals, or correctness indicators.

Oracle upper bound
The accuracy achievable by a hypothetical policy with offline access to gold labels, used only as a diagnostic ceiling. It is not a baseline method.

Provider-specific evidence
Results obtained from a particular provider–model pairing (here, Cohere command-r-plus-08-2024). Claims are explicitly scoped to this setting and do not generalize beyond it without additional replication.

3.3 Method

The main method, the **Failure-Trace Allocator (FTA)**, is an answer-selection layer applied *after* candidate generation (Figure 1). It does not generate new candidates and does not issue additional inference calls: it selects a final answer from already-produced frontier and external outputs under the same per-method budget cap. Its two gates—the *Low-Depth Guard* and the *External-Consensus Fallback*—fire in strict priority order; at most one gate acts per example. All runtime decisions are gold-free: no gold answer, exact-match signal, or correctness label is accessed. Implementation-name mapping is retained in Table 16.

At each example, FTA consumes three classes of runtime-legal inputs: the frontier answer candidate, the three evaluated external-baseline answers, and failure-trace metadata emitted during normal frontier execution. The output is a single final answer plus an explicit policy outcome (`fix2`, `fix4`, or frontier retained), so every decision can be audited after the run using only logged runtime fields. This interface is intentionally narrow: the main method does not depend on learned scores or post-hoc labels.

From an intelligent-systems perspective, FTA can be read as an instance-level selective deferral policy over a fixed candidate pool: rather than learning a new generator, it uses runtime-visible failure-trace and agreement signals to decide whether to retain the frontier answer or defer to an external-consensus answer under a matched-budget inference contract. This places the method in the same conceptual space as selective prediction, dynamic ensemble selection, and cost-aware LLM routing, while keeping the decision rule deterministic and auditable.

Frontier generator and runtime metadata. The frontier method (Section 3.2) produces each candidate by exploring multiple reasoning *branches*, where a branch is one candidate trajectory and its *depth* is the number of expansion steps from the root. As a side-effect of normal inference, the frontier logs four runtime-visible metadata fields that FTA consumes. `frontier_support` is the integer count of branches that converge on the selected answer; a value of zero or one indicates a low-support, single-branch selection. `candidate_pool_answer_group_count` is the integer count of distinct answer equivalence classes across all branches explored, with higher values indicating greater internal disagreement. `override_reason` is a string set by the frontier's internal selection step; the values `"single_weak_frontier_branch"` (at most one branch supports the selected answer) and `"direct_frontier_agree"` (the frontier answer matches the frontier's own internal reference selection, indicating internal self-consistency) are the two states relevant to the LDG and ECO gates respectively. `direct_frontier_agree` is a boolean indicator of the same internal-consistency condition. All fields are recorded during ordinary inference without gold-label access; FTA uses them exactly as logged, making the decision boundary auditable from the run records.

Conceptually, FTA uses three abstract predicates. $\textsc{WeakFrontier}(x)$ is true when the frontier trace indicates low support or a single-branch selection. $\textsc{ExternalMajority}(x)$ returns an external answer when at least two of L1, S1, and TALE agree, with a deterministic fallback only when all three differ. $\textsc{ExternalConsensusDisagree}(x)$ is true when L1, S1, and TALE unanimously agree with one another and disagree with the frontier answer. The exact field-level realizations of these predicates are given in Table 1 and Table 6; the raw log-field names are implementation details documented in the reproducibility appendix.

The *Low-Depth Guard* is motivated by a structural reliability pattern. When the frontier search terminates with a single weak branch—signalled by logged branch-trace metadata—the resulting candidate is treated as evidence of under-exploration. In such cases, routing to an external majority answer provides an independent cross-policy signal without spending extra compute. The *External-Consensus Fallback* addresses a complementary pattern: if the frontier's selection is internally self-consistent yet all three external methods unanimously prefer a different answer, that unanimous disagreement is treated as high-confidence evidence that the frontier selection may be wrong. Partial external agreement (one or two externals disagreeing) is intentionally not used in the main method because it did not provide validation-grade evidence in the evaluation setting.

Selective-recovery proposition. Let S be a runtime-identifiable subset of examples. Let $e_F(S)$ be frontier error rate on S and $e_M(S)$ be external-majority error rate on S. If $e_F(S) > e_M(S)$, then replacing frontier with external majority on S improves expected accuracy by

$$P(S)\left(e_F(S) - e_M(S)\right).$$

The Low-Depth Guard (LDG) and External-Consensus Fallback (ECO) are gold-free empirical heuristics intended to identify such subsets without using correctness labels at runtime.

The control flow follows a strict precedence chain (Figure 1, Table 6). The Low-Depth Guard is checked first; when it fires, the policy routes to an external majority answer and stops. This priority reflects the interpretation of the weak-frontier trace as a generation-quality warning: when exploration appears narrow, the policy does not additionally check external consensus. If LDG does not fire, the External-Consensus Fallback is checked; when that condition fires, the policy routes to the external unanimity answer. If neither gate fires, the frontier answer is retained. The default therefore preserves the original frontier method on all examples where neither high-confidence failure trace is present. Conflict handling is deterministic: at most one gate can act, and external majority ties are resolved by the fixed priority order documented in Table 6. That fallback is retained for reproducibility rather than claimed as optimal; the sensitivity noted in Table 14 shows that alternative tie orders change Aggregate-720 replay by less than one percentage point.

All trigger features are runtime-legal and gold-free by construction. They are computed only from model outputs and frontier trace metadata that are already visible at inference time, never from gold answers, exact-match signals, example identifiers, or artifact paths. Table 15 gives the complete legality inventory, including allowed inputs and explicitly excluded fields. This legality boundary is central to the method definition, not only a reporting constraint.

Relative to L1, S1, TALE, and the frontier baseline, FTA plays a different role in the stack. L1, S1, TALE, and frontier each provide candidate answers under the same matched per-method budget; FTA then decides whether to keep the frontier answer or override it using external agreement signals. This separation between candidate generation and final allocation is what allows the policy to improve selection quality without additional model calls. Observed gains come from switching the final answer on a small trace-identified subset of examples rather than from increasing inference budget.

3.4 Low-Depth Guard (LDG)

The Low-Depth Guard detects frontier answers arising from structurally weak search branches and replaces them with an external majority vote. The trigger relies exclusively on fields logged during the frontier run: the override-reason indicator and the candidate-pool group count.

3.5 External-Consensus Fallback (ECO)

The External-Consensus Fallback identifies examples where the frontier agreed with its own internal reserve selection, yet all three external baselines unanimously chose a different answer. When this unanimous disagreement holds, the policy substitutes the external consensus answer.

3.6 Combined Policy Summary

Table 1 gives the complete pseudocode using exact runtime field names. Parameter settings and implementation-name mapping are documented in the appendix reproducibility tables.

The exact field-level trigger conditions are also tabulated in Table 6 for reproducibility.

Table 1: Failure-Trace Allocator (FTA) combined policy. All inputs are gold-free runtime fields. Named constants and name mapping are documented in Table 16 and Table 6.

Step	Operation
Inputs per example	
	`frontier_answer`, `result_metadata` (from frontier run logs)
	`l1_answer`, `s1_answer`, `tale_answer` (external baseline outputs)
	`override_reason`, `frontier_support` $\in \mathbb{Z}_{\geq 0}$, `candidate_pool_answer_group_count` $\in \mathbb{Z}_{>0}$
Step 1 — Low-Depth Guard (LDG)	
1a	**if** `override_reason` contains `"single_weak_frontier_branch"`
	or (`frontier_support` $= 0$ **and** `candidate_pool_answer_group_count` ≥ 2)
	then:
1b	Compute `ext` $\leftarrow$ majority answer among {`l1`, `s1`, `tale`} (at least 2 agree);
	if no majority, use priority order TALE $\succ$ S1 $\succ$ L1
1c	**if** `ext` exists **and** `ext` $\neq$ `frontier_answer`:
	return `ext`, `policy` $\leftarrow$ `fix2` *(guard fires; stop)*
Step 2 — External-Consensus Fallback (ECO)	
2a	**if** `override_reason` $=$ `"direct_frontier_agree"`
	and `l1_answer` $=$ `s1_answer` $=$ `tale_answer`
	and `l1_answer` $\neq$ `frontier_answer`
	then:
2b	**return** `l1_answer`, `policy` $\leftarrow$ `fix4` *(consensus fallback fires; stop)*
Step 3 — Default	
3	**return** `frontier_answer`, `policy` $\leftarrow$ `original`
Output	
	Selected answer and `policy` label; no gold labels accessed at any step.

4 Experimental Setup

4.1 Data, model, and protocol

All retained comparisons are conducted on **Cohere command-r-plus-08-2024** with a fixed per-method budget cap of $B = 6$ logical API calls per example. The dataset is `openai/gsm8k`. Evaluation conditions are matched across all compared candidate-producing methods: every method operates under the same budget cap, provider, and model, so accuracy differences are not confounded by unequal per-method compute.

Matched-budget definition. A comparison is *matched-budget* if and only if every compared candidate-producing method is subject to the same maximum logical API call count B per example, with the same provider and model. No method in the retained comparison receives extra calls or a different model. FTA is then applied as a post-generation selection policy over the logged candidate answers and frontier metadata. This contract supports clean method-to-method comparison, but it does not imply total compute parity with a single baseline run when multiple candidate-producing methods are executed together.

4.2 Cost and resource usage

Table 8 reports per-method resource usage measured from logged Aggregate-720 runs on Cohere command-r-plus-08-2024. Reported metrics include average logical API calls, average input tokens, average output tokens, and average latency in seconds. Total tokens (input + output) can be derived by summing the respective columns. Budget parameter $B = 6$ is shown for completeness.

FTA adds negligible selection overhead once candidate answers and frontier metadata are available, but the main deployment cost is candidate-pool construction. If a deployment must generate frontier, L1, S1, and TALE outputs from scratch, total wall-clock and API cost depends on which candidate producers are executed and may exceed a single method's $B = 6$ run. In other words, FTA is a zero-extra-call selector after generation, not a claim of one-run compute equivalence to all baselines combined. In the logged Aggregate-720 runs, the frontier method has higher average latency than L1, S1, or TALE. We report cost in logical calls, tokens, and latency; monetary cost depends on provider pricing and is not used as the primary comparison metric. Dollar-normalized cost analysis, latency percentile distributions, and multi-provider token cost benchmarks are identified as future work (Section 6.1).

4.3 Parameter grounding

Table 14 (Appendix) documents the key parameters and design choices used in the two-gate FTA policy, including their values, roles, and the robustness evidence available. We report parameter grounding and robustness checks rather than a full continuous hyperparameter sweep; a full threshold sweep is left for future work.

4.4 Primary splits and disjointness

The primary aggregate (Aggregate-720) is formed from three disjoint sources: seed 41 (300 examples), seed 61 (120 examples), and seed 71 (300 examples). Cross-source overlap checks confirm zero shared example IDs and zero shared question hashes, ensuring that the aggregate evidence is drawn from non-overlapping subpopulations.

4.5 Statistics

We report paired bootstrap confidence intervals [22] (n = 5,000 resamples) on Final-300 and source-stratified aggregate confidence intervals on Aggregate-720. The bootstrap resample count of 5,000 is standard for stable CI estimation at the sample sizes used. All confidence intervals are two-sided at the 95% level. When shown, the *bootstrap positive-delta fraction* is the fraction of bootstrap resamples in which the FTA delta is positive; it is a descriptive bootstrap summary and should not be interpreted as a classical frequentist p-value.

5 Results and Analysis

Main results are listed in Table 2 and Figure 2. FTA is the main method; all other evaluated variants were not retained (see Section 5.1 and Table 10).

Final-300. On the unbiased 300-example evaluation (seed 71, budget 6), FTA reaches **86.67%** (260/300), versus L1 83.00% (249/300), S1 82.00% (246/300), and TALE 78.33% (235/300). Paired bootstrap 95% confidence intervals (5 000 resamples) for FTA minus each baseline on this split:

- vs. L1: +3.67 pp, 95% CI [+0.33, +7.00], bootstrap positive-delta fraction 0.983
- vs. S1: +4.67 pp, 95% CI [+1.00, +8.33], bootstrap positive-delta fraction 0.993
- vs. TALE: +8.33 pp, 95% CI [+5.00, +12.00], bootstrap positive-delta fraction 1.000

All three CI lower bounds are strictly positive on this split.

Aggregate-720 (primary). The primary claim rests on the disjoint Aggregate-720 evaluation combining three independent sources (seeds 41, 61, 71; comprising 300, 120, and 300 examples respectively, with no example-id or question-hash overlap). FTA reaches **80.69%** (581/720), versus L1 77.64% (559/720), S1 77.08% (555/720), and TALE 75.14% (541/720). Source-stratified bootstrap 95% confidence intervals (5 000 resamples):

- vs. L1: +3.06 pp, 95% CI [+0.83, +5.42], bootstrap positive-delta fraction 0.995
- vs. S1: +3.61 pp, 95% CI [+1.39, +5.69], bootstrap positive-delta fraction 0.999
- vs. TALE: +5.56 pp, 95% CI [+3.61, +7.50], bootstrap positive-delta fraction 1.000

- vs. source-local best external: +3.06 pp, 95% CI [+1.11, +5.00], bootstrap positive-delta fraction 0.999

All aggregate source-stratified CI lower bounds are strictly above zero. The margin vs. best external exceeds 1 pp even at the lower CI bound.

Pooled ensemble baseline. We also evaluate a simple pooled answer ensemble over the four already-generated answers: frontier, L1, S1, and TALE (Table 4). The primary pooled baseline uses strict majority vote over the four answers and falls back to the frontier answer for 2–2 ties or all-different cases; this tie-breaker is deterministic and gold-free. On Final-300, this Pooled-4 baseline reaches 84.33% (253/300), while FTA reaches 86.67% (260/300), a +2.33 pp difference with paired bootstrap 95% CI [−0.67, +5.67]. On Aggregate-720, Pooled-4 reaches 79.86% (575/720), while FTA reaches 80.69% (581/720), a +0.83 pp difference with source-stratified 95% CI [−1.11, +2.78]. An external-only L1/S1/TALE majority baseline is even closer: 85.33% on Final-300 and 80.28% on Aggregate-720. Thus, FTA remains ahead by point estimate, but the pooled-ensemble comparison is not statistically separated and is treated as a robustness check rather than a retained superiority claim. The external fallback order used by LDG is deterministic and fixed for reproducibility; it is not presented as an optimized learned ranker. A replay sensitivity check over alternative external tie orders changes the Aggregate-720 LDG-only accuracy from 80.00% to 80.42%, with frontier fallback on all-different external ties reaching 80.83%; these variants are reported only as sensitivity because they were not the main method.

FTA vs. External-3 majority: selective vs. uniform override. The External-3 majority baseline (L1/S1/TALE vote) reaches 85.33% on Final-300 and 80.28% on Aggregate-720, placing it close to FTA by point estimate. This proximity merits interpretation, because the two approaches differ in operating principle: External-3 applies a uniform vote over the three external baselines on every example, while FTA retains the frontier answer on 593 of 720 Aggregate-720 examples and overrides only when runtime failure-trace metadata indicates structural weakness (LDG) or unanimous high-confidence external disagreement (ECO). FTA's value is therefore not a pooled-vote point estimate but a transparent, auditable selective-deferral rule in which each override is attributable to a named gate and a logged trigger field from the frontier run. The ECO component contributes only five triggers on Aggregate-720—all recovering from incorrect frontier answers—and should be read as a low-frequency high-precision safeguard rather than a high-coverage source of aggregate recovery. In this evaluation, FTA and External-3 reach similar aggregate accuracy through different mechanisms: FTA preserves the frontier signal on most examples and defers selectively, while External-3 discards the frontier uniformly.

Source-by-source breakdown. Table 3 gives the per-source accuracy for all evaluated methods. Estimates are consistent across sources: FTA leads by point estimate in every individual source split, with seed 41 showing the smallest advantage (83.33% vs. 80.33% for all three baselines) and seed 71 showing the largest (86.67% vs. 83.00% for L1). The seed 61 source is substantially harder in absolute terms: frontier, L1, S1, and TALE all fall to 57.50%, 57.50%, 56.67%, and 54.17%, respectively. The provider/model/budget configuration is unchanged, and the mean question length is only modestly higher than the other sources (48.5 words vs. 46.1 for seed 41 and 44.6 for seed 71). The available evidence therefore points to source-sample difficulty rather than a method-specific run configuration change. FTA still leads the best external on seed 61 by +1.67 pp, and the source-stratified aggregate confidence interval accounts for this source variation.

Win/loss/tie decomposition. Appendix Table 12 gives the per-example win/loss/tie counts versus each baseline on Final-300. FTA wins on 19 examples vs. L1 (losses: 8, ties: 273), 23 examples vs. S1 (losses: 9, ties: 268), and 27 examples vs. TALE (losses: 2, ties: 271). The

Table 2: Main validated results (final300 and aggregate720). Deltas are percentage points vs baselines on aggregate720.

Policy	Final300 Acc.	Final300 c/n	Agg720 Acc.	Agg720 c/n	ΔvsL1	ΔvsS1	ΔvsTALE	Promoted
frontier original	76.67	230/300	75.28	542/720	−2.36	−1.81	+0.14	no
Low-Depth Guard (LDG)	85.67	257/300	80.00	576/720	+2.36	+2.92	+4.86	no
Failure-Trace Allocator (FTA)	**86.67**	**260/300**	**80.69**	**581/720**	**+3.06**	**+3.61**	**+5.56**	**yes**
FIX-5	78.67	236/300	75.28	542/720	−2.36	−1.81	+0.14	no
Pooled-4 ensemble	84.33	253/300	79.86	575/720	+2.22	+2.78	+4.72	no
External-3 ensemble	85.33	256/300	80.28	578/720	+2.64	+3.19	+5.14	no
L1	83.00	249/300	77.64	559/720	0.00	+0.56	+2.50	no
S1	82.00	246/300	77.08	555/720	−0.56	0.00	+1.94	no
TALE	78.33	235/300	75.14	541/720	−2.50	−1.94	0.00	no
best external	83.00	249/300 (L1)	77.64	559/720 (L1)	0.00	+0.56	+2.50	no

Table 3: Source-by-source accuracy (%) and aggregate context. Sensitivity source (seed 31) is a labeled auxiliary set, not part of retained Aggregate720 evidence. Bold denotes the best value within each source row (ties allowed).

Source	Seed	N	Frontier	LDG	FTA	L1	S1	TALE	Δ(FTA vs best ext.)
main_300_seed41	41	300	81.00	82.67	**83.33**	80.33	80.33	80.33	+3.00
independent_120_seed61	61	120	57.50	**59.17**	**59.17**	57.50	56.67	54.17	+1.67
final_300_seed71	71	300	76.67	85.67	**86.67**	83.00	82.00	78.33	+3.67
sensitivity_100_seed31	31	100	73.00	80.00	**82.00**	76.00	77.00	**82.00**	0.00
Aggregate720 (primary)	41+61+71	720	75.28	80.00	**80.69**	77.64	77.08	75.14	+**3.06**

extremely low loss counts vs. TALE confirm that the policy's recoveries substantially outweigh its regressions on this evaluation.

Gate-action decomposition. Table 7 decomposes the main method into gate actions. Across Aggregate-720, LDG fires on 122 examples, ECO fires on 5 examples where LDG did not fire, and no gate fires on 593 examples. The ECO actions are rare but highly precise: before ECO fires, the frontier answer is incorrect on all five triggered examples; after ECO fires, all five are correct. The same pattern holds on Final-300, where ECO fires on three examples and contributes three wins, zero losses, and zero ties relative to the frontier answer and to LDG-only replay. Accordingly, ECO should be interpreted as a low-frequency high-precision safeguard in this evaluation, not as a standalone high-coverage component with independent statistical reliability. Most of the aggregate recovery mass comes from LDG.

Sensitivity set (supporting diagnostic only). Table 3 includes a row for a 100-example set (seed 31, labeled "sensitivity-only") that was used as a development diagnostic during early policy calibration. Because this set was involved in parameter selection for earlier evaluated variants, it is excluded from the primary Aggregate-720 evidence and from the bootstrap confidence intervals reported above. On this set, FTA achieves 82.00%, tied with TALE and ahead of L1 (76.00%) and S1 (77.00%). When the sensitivity set is added to the aggregate ($N = 820$), the positive point-estimate advantage of FTA over all three external baselines is preserved, and the overall picture is unchanged. All retained accuracy claims in this paper are based on Final-300 and Aggregate-720 only.

5.1 Ablation Study and Not-Retained Variants

We evaluate ablation and extension variants to test whether the retained two-gate policy can be improved by additional routing, extra-action, clustering, parsing, or pool-miss logic. Table 5 summarizes the evaluated variants from FIX-1 through FIX-9. All variants were evaluated via postrun replay on existing candidate artifacts; no additional inference calls were issued for ablation purposes. None of the variants met retention criteria beyond FTA, and they are reported to delimit the supported claim.

Table 4: Pooled answer-ensemble baselines computed from the same per-example frontier, L1, S1, and TALE outputs. Pooled-4 uses a strict majority over four answers and falls back to the frontier answer on ties or all-different cases. External-3 uses majority vote over L1/S1/TALE with the fixed, deterministic TALE≻S1≻L1 tie fallback used by the main method; tie sensitivity is summarized in Table 14. Deltas and win/loss/tie counts compare Failure-Trace Allocator (FTA) to the pooled baseline.

Split	FTA	Pooled-4	External-3	Δ vs Pooled-4	W/L/T vs Pooled-4	95% CI
Final-300	260/300 (86.67%)	253/300 (84.33%)	256/300 (85.33%)	+2.33 pp	16/9/275	[−0.67, +5.67]
Aggregate-720	581/720 (80.69%)	575/720 (79.86%)	578/720 (80.28%)	+0.83 pp	28/22/670	[−1.11, +2.78]

Table 5: Ablation variants and non-main method outcomes. Bold denotes the strongest retained-policy result in the main comparison block.

Policy	Target failure mode	Accuracy / delta summary	Retained
frontier original	baseline reference	final300=76.67%; agg720=75.28%	no
FIX-1 support-aware	tie / present-not-selected	77.00% (delta vs frontier +4.00 pp)	no
Low-Depth Guard (LDG, implementation: FIX-2) low-depth guard	single-weak-frontier-branch / low depth	80.00%	no (superseded by Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4))
FIX-3 calibration	score-absent non-weak-frontier edge cases	73.00% (applied 0 cases)	no
External-Consensus Fallback (ECO, implementation: FIX-4) consensus gate	direct-frontier-agree + unanimous external disagreement	75.00% alone; 82.00% with Low-Depth Guard (LDG, implementation: FIX-2)	part of Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4)
Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4)	low-depth + external-consensus	final300=86.67%; agg720=80.69%	**yes**
FIX-1+2+3+4	combined heuristic stack	81.00%	no
FIX-5 TALE-default router	agreement-region switching	overnight300=80.33%; final300=78.67%	no
FIX-6 extra-action variant	residual errors needing extra action	net vs Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4): frontier −1, TALE −4	no
FIX-7 cluster selector	cluster-level reranking / parser guards	best-rule local delta vs Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4): +0.67 pp	no
FIX-8 robust parser	parser/canonicalization errors	final300 delta +0.00 pp; agg720 +0.00 pp	no
FIX-9 pool-miss detector	generation-limited residuals	23/24 all-methods wrong generation-likely; 0 rule opportunities	no

Retained gate components. LDG alone achieves 80.00% on Aggregate-720 (+2.36 pp over the frontier baseline), confirming that the Low-Depth Guard carries most of the combined gain. ECO alone achieves 75.00% on Final-300—below LDG—but contributes an additional +0.69 pp when combined with LDG, reflecting its role in recovering cases where LDG does not fire. FIX-3 (calibration on score-absent non-weak-frontier edge cases) was evaluated but applied to zero examples in the primary splits, contributing no accuracy change; it is excluded from the combined policy.

Non-retained variants. FIX-5 (TALE-default router) produced zero switches on seeds 41 and 61, equalling TALE performance on both splits, and only a marginal net positive switch on seed 71 (+1 example over TALE, Final-300 78.67% vs. 78.33%). This outcome is −8.00 pp below FTA on the same split. Note that the Aggregate-720 accuracy for FIX-5 (75.28%) numerically coincides with that of the frontier baseline; this reflects the zero-switch behavior on seeds 41 and 61, not an equivalence by design. The TALE-default design assumed a reliable agreement region that did not materialise consistently across evaluation seeds; see Section 6 for interpretation. FIX-6 (extra-action variant) showed a nonnegative pilot signal in the initial 40-case overlapping evaluation, but an independent 80-case follow-up yielded net −1 for extra frontier actions and net −4 for extra TALE retries relative to FTA. Early-stage signal that fails to replicate on disjoint data is the standard not-retained criterion. FIX-7 (cluster-level selector) achieved at most +0.67 pp in the best rule variant on Final-300, below the threshold for retention in the main method. FIX-8 (robust parser/canonicaliser) produced zero accuracy-changing switches on both Final-300 and Aggregate-720, indicating that parse-failure residuals are rare in this evaluation setting. FIX-9 (pool-miss detector) found 23 of 24 all-methods-wrong residuals to be generation-limited, with no selector-only remedy available; see the failure analysis in Section 5.2.

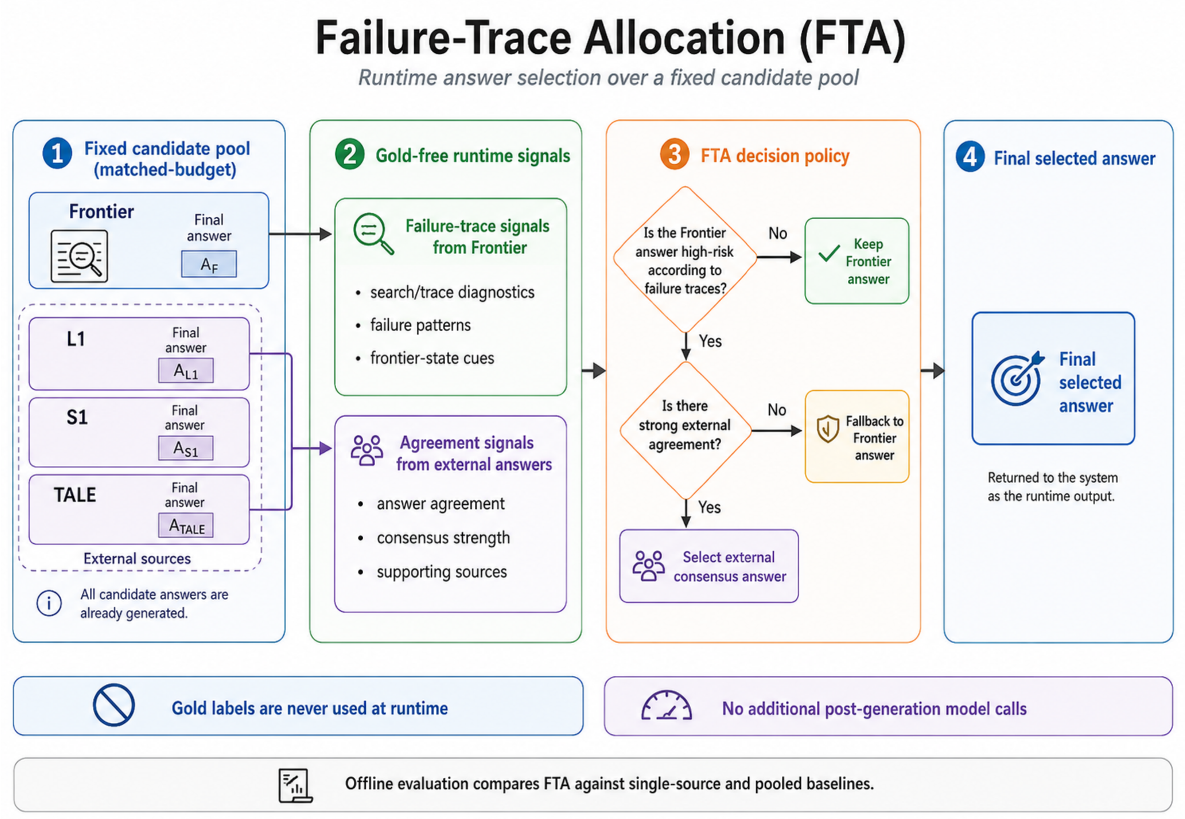

Figure 1: Failure-Trace Allocator (FTA) decision pipeline. Step 1: a fixed candidate pool is pre-generated at matched budget (Frontier, L1, S1, TALE; all candidate answers are generated before runtime selection begins). Step 2: gold-free runtime signals are derived from the frontier trace (failure diagnostics, failure patterns, frontier-state cues) and from agreement signals across external answers. Step 3: the FTA decision policy applies the two-gate logic described in Section 3.3: the Low-Depth Guard (LDG) checks for a structurally weak frontier trace; the External-Consensus Fallback (ECO) checks for unanimous external disagreement with the frontier answer. Step 4: the selected answer is returned as the runtime output. Gold labels are never used at runtime; no additional post-generation model calls are issued. Offline evaluation compares FTA against single-source and pooled baselines.

5.2 Failure Analysis

Table 9 and Figure 4 classify the 40 residual errors in the Final-300 evaluation by root cause. Three categories emerge. *All-methods-wrong* cases (24/40, 60.0%) are examples where every evaluated method—the frontier generator and all three external baselines—produced an incorrect answer. Because no correct answer appears in the candidate pool, no selection policy can recover these cases; they represent a hard ceiling for the current approach. *Parser/canonicalization failures* (12/40, 30.0%) are cases where the correct numerical answer was present in the pool but not extracted or matched, due to format mismatches between the model's output format and the evaluation matcher. *Present-not-selected* cases (4/40, 10.0%) are cases where the correct answer was available in the pool but the FTA selection policy did not surface it.

Within the all-methods-wrong category, the dominant subtype is *multi-step composition traps* (10/24), problems where the error is embedded in intermediate reasoning rather than in the final answer selection step. Near-miss numeric errors (5/24), different-wrong-answer divergence with no correct cluster (4/24), and false consensus (2/24)—where all methods agree on the same incorrect answer—account for most of the remainder. False-consensus cases are the most

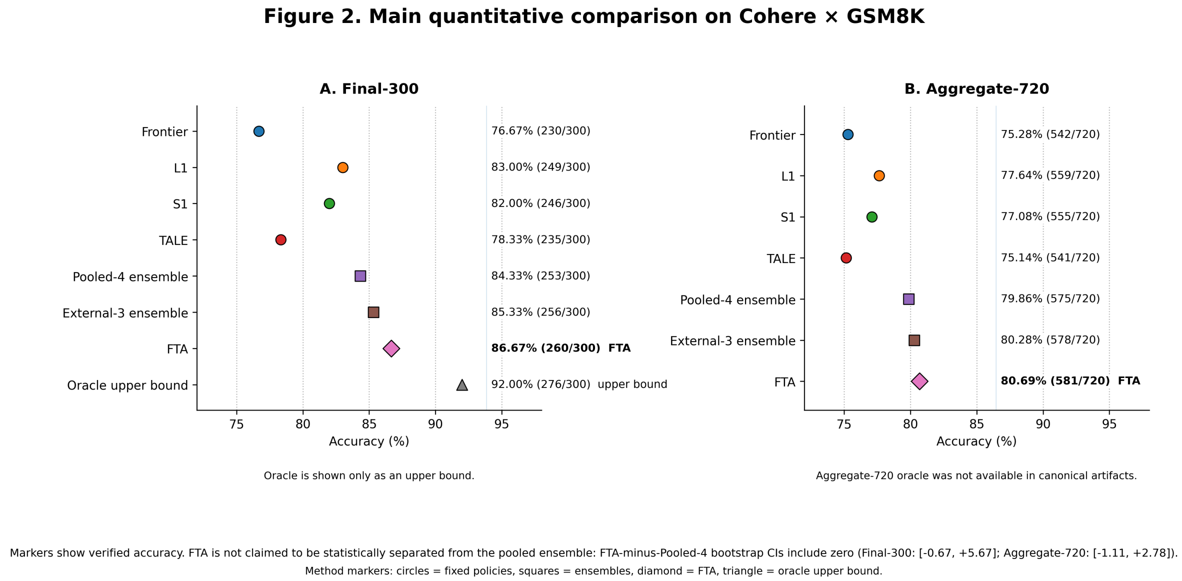

Figure 2: Main quantitative comparison on Cohere × GSM8K. Panel A shows Final-300 and Panel B shows Aggregate-720. The oracle is shown only as an upper bound, not as a deployable baseline. FTA is not claimed to be statistically separated from the pooled ensemble because the FTA-minus-Pooled-4 bootstrap confidence intervals include zero.

Table 6: Exact policy rule definitions for reproducibility. Abstract predicates are implemented by gold-free runtime fields only.

Component	Predicate / exact condition	Action	Gold-free rationale
Low-Depth Guard (LDG): WeakFrontier	`override_reason` contains **`single_weak_frontier_branch`** **or** (`frontier_support`= 0 **and** `pool_group_count`$\geq$ 2)	Fallback to external majority vote; if no majority use priority TALE≻S1≻L1; else keep frontier	Uses only runtime metadata and baseline answers; no gold/exact labels
External-Consensus Fallback (ECO): ExternalConsensusDisagree	`override_reason`=`direct_frontier_agree` **and** `l1`=`s1`=`tale` **and** unanimous external $\neq$ frontier	Replace frontier with the unanimous external answer	Uses only runtime override reason and method answers
Combined Failure-Trace Allocator (FTA)	If LDG fires: apply LDG and stop; else if ECO fires: apply ECO; else keep frontier	Single-fire precedence chain	No correctness labels in trigger chain

resistant to any selection-based intervention: unanimous agreement provides no gating signal when the consensus itself is wrong.

These findings directly inform the scope of FTA and the directions for future work. The all-methods-wrong class (60% of failures) is generation-limited: addressing it requires either stronger base generators or training-time interventions that improve coverage of hard compositional patterns, not better post-hoc selection policies. The parser/canonicalization class (30% of failures) is potentially tractable with improved answer normalization; however, the FIX-8 prototype (robust parser/canonicaliser; see Table 5) produced zero accuracy-changing switches in this setting, suggesting the problem is subtler than simple string-matching improvements. The present-not-selected class (10% of failures) represents the only residual opportunity that a selection policy could in principle address without improved candidate generation.

6 Discussion

Why failure-trace gating is effective. The two gates in FTA target failure modes that are identifiable at runtime without any access to gold labels. The Low-Depth Guard (LDG)

Table 7: Gate-action decomposition for the retained FTA policy. ECO fires only after LDG does not fire. The final columns report correctness before and after ECO on the examples where ECO fires, relative to the original frontier answer.

Split	N	LDG fires	ECO fires	No gate	ECO before	ECO after	ECO W/L/T
seed41	300	38	2	260	0/2	2/2	2/0/0
seed61	120	21	0	99	0/0	0/0	0/0/0
seed71 (Final-300)	300	63	3	234	0/3	3/3	3/0/0
Aggregate-720	720	122	5	593	0/5	5/5	5/0/0

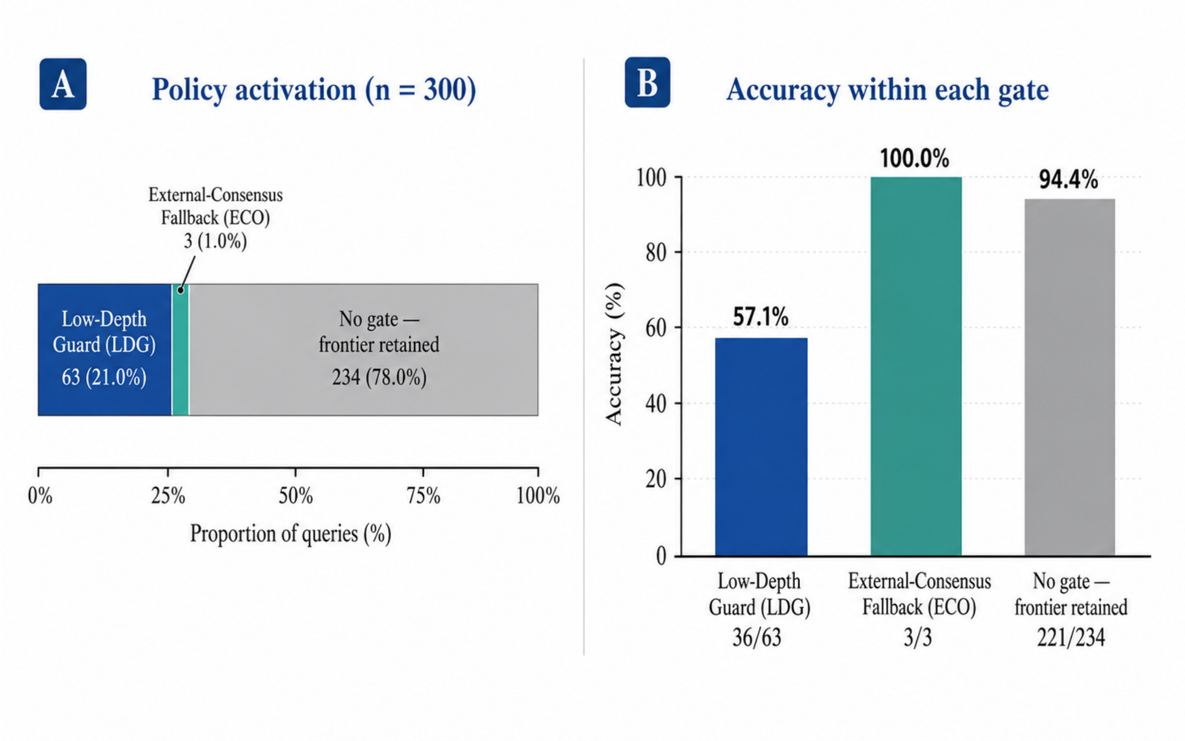

Figure 3: FTA policy activation on Final-300. Panel A shows the gate composition: Low-Depth Guard (LDG), External-Consensus Fallback (ECO), and no-gate cases. Panel B reports the within-gate accuracy for each case type.

addresses a structural weakness: when the frontier search terminates with a single weak branch—signalled by the logged override-reason field—the resulting answer is drawn from a narrow slice of the candidate space. In such cases, the external majority vote is empirically more reliable. The External-Consensus Fallback (ECO) targets a complementary pattern: when the frontier selected its answer with high internal confidence, yet all three independent external baselines unanimously prefer a different answer, that unanimous disagreement provides strong evidence that the frontier selection is wrong. Both signals are available as logged metadata from normal frontier execution; no additional inference or labelling is required.

When FTA helps and when it does not. FTA provides the greatest benefit on examples where the frontier's failure mode is structurally signalled—narrow search depth or unanimous external disagreement—because these are precisely the cases the two gates are designed to detect. In the Final-300 evaluation, LDG fires on 63 of 300 examples (21.0%) and yields 36 wins against 9 losses versus the frontier, while ECO fires on only 3 examples (1.0%) with zero losses. The no-gate majority (234/300, 78.0%) retains the frontier answer with 94.4% accuracy, indicating

Table 8: Aggregate720 per-method cost and resource summaries from logged runs. Provider: Cohere command-r-plus-08-2024; each candidate-producing method is evaluated with budget cap $B = 6$ (matched-budget condition). Avg total tokens = avg input + avg output. Monetary cost is not reported; it depends on provider pricing (see Section 4.2).

Method	Avg logical calls	Avg input tok.	Avg output tok.	Avg total tok.	Avg latency (s)	Budget
Frontier	2.60	1153.97	229.94	1383.91	13.30	6
L1	1.21	515.87	114.14	630.01	4.11	6
S1	2.62	1110.20	160.50	1270.70	5.15	6
TALE	1.44	612.38	117.90	730.28	4.63	6

Table 9: Failure taxonomy for final300 residual errors.

Failure group	Count	Share	Likely fix class
final300 total failures	40	1.000	mixed
all methods wrong	24	0.600	new candidate generation likely
parser / canonicalization issue	12	0.300	parser / canonicalization
present not selected	4	0.100	selector
Subtypes of all-methods-wrong			
multi-step composition trap	10	0.250	selector-only unlikely
near-miss numeric error	5	0.125	selector-only unlikely
different wrong answers, no correct cluster	4	0.100	mixed (rare selector possible)
same wrong answer, false consensus	2	0.050	selector-only unlikely
shared unit/rate trap	2	0.050	selector-only unlikely
quantity selection trap	1	0.025	selector-only unlikely

that the frontier is reliable on most examples and the policy correctly defers to it.

FTA does not help—and cannot—when all methods in the candidate pool produce incorrect answers for the same example. This *pool-miss* failure class accounts for 24 of the 40 residual errors on Final-300 (60%), representing an irreducible floor for any post-generation selection policy operating over the current pool. Additionally, the ECO gate requires unanimous external agreement, so cases where the external baselines partially but not unanimously disagree with the frontier are handled by the frontier default. Extending the method to handle partial agreement would require stronger statistical justification, as such cases showed more ambiguous outcomes in development.

The current evaluation is scoped to GSM8K under the Cohere command-r-plus-08-2024 setting. Preliminary evidence on other providers (Section 6.1) suggests the failure-trace signal may have broader utility, but the main findings should not be extrapolated to other benchmarks or providers without independent replication.

Not-retained variants. Several algorithmic extensions were developed and evaluated during the research but did not meet validation-grade evidence criteria (Table 10). A TALE-default router produced zero net switching effect across evaluation seeds, equalling TALE performance exactly on seeds 41 and 61, because the assumed agreement region did not materialise consistently. An extra-action variant—collecting additional frontier or retry actions for predicted failure cases—showed a non-negative initial signal in a 40-case pilot but failed to replicate on a disjoint 80-case follow-up (net -1 for extra frontier actions; net -4 for TALE retries), suggesting the pilot signal was run-specific rather than generalisable. A cluster-level selector yielded only limited offline gain without sufficient aggregate validation. These outcomes indicate that rule-based post-generation selection gains within the current pool composition are likely close to saturation, and further improvements will require better candidate generation or learned allocation policies.

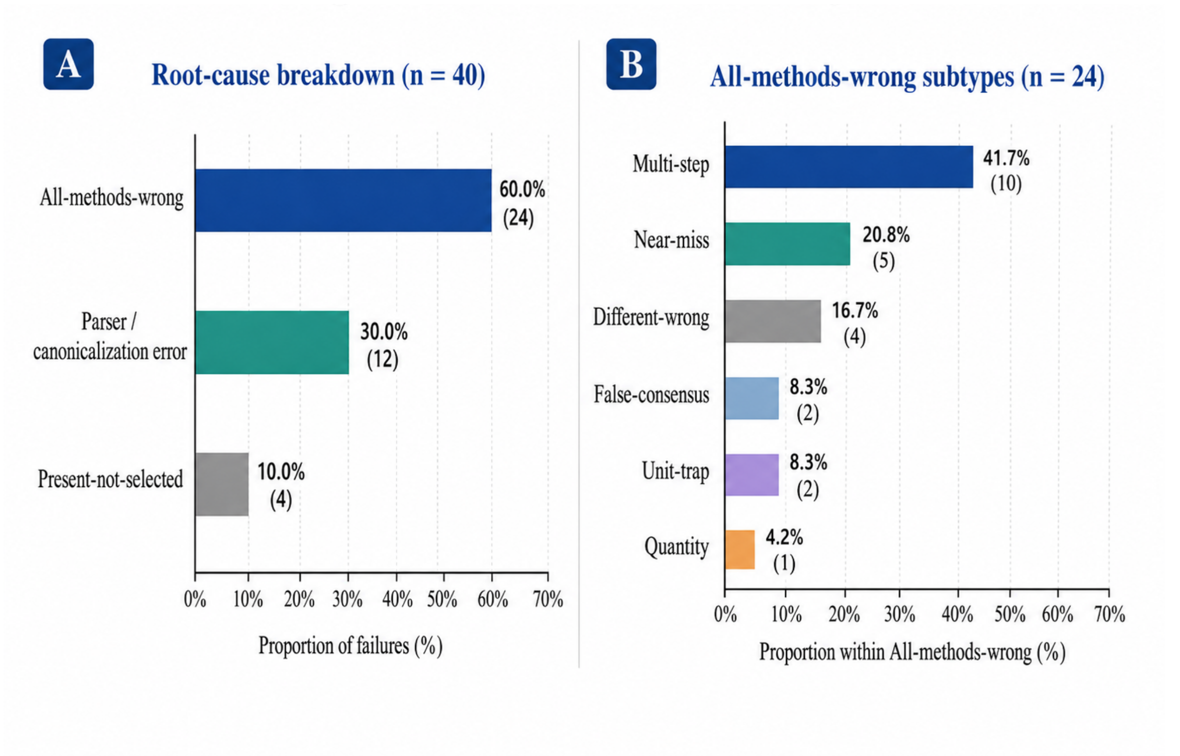

Figure 4: Residual failure analysis after FTA. Panel A shows the repository-verified three-category breakdown over the 40 remaining Final-300 failures. Panel B decomposes the all-methods-wrong subset into its verified subtypes.

Evidence from the Aggregate-720 evaluation. The most substantive evidence shift in this work is the transition from a single-run 300-example evaluation (seed 41, where the CI lower bound vs. L1 crossed zero at $[-0.48, +5.71]$) to the disjoint Aggregate-720 evaluation (seeds 41, 61, 71). By combining three independently sampled evaluation splits, the source-stratified bootstrap CI narrows sufficiently that the lower bound versus every external baseline is strictly positive. The source-stratified design controls for seed-specific variance, and the zero-overlap check on example-id and question-hash confirms independence. The point-estimate advantage of FTA is consistent across all three seeds, which is the primary reason the aggregate CIs no longer include zero.

Multi-provider supporting evidence. As supporting evidence for the generality of the failure-trace signal, a preliminary multi-provider experiment (D9) trains a learned gate on the same failure-trace features across three providers. Cross-validated accuracy on the D9 gate reaches 0.5018 ± 0.0252 against a frontier baseline of 0.3436, with zero false-override events across all tested providers and decision thresholds. The Cloudrift/Qwen provider contributes the strongest rescue signal (frontier 37.5%, D6-gated 55.0%, 24 unique-correct outcomes, zero regressions with lenient extraction). These results are consistent with the hypothesis that frontier structural weakness is a provider-independent signal, but the evaluation scope is preliminary (80–160 examples per provider); the findings constitute supporting evidence, not a second primary claim. Broader multi-provider evaluation at matched scale is a natural next step.

Evidence scope and generalization. The main findings are scoped to the tested evaluation setting. What this evidence does *not* establish is generalization beyond that scope: whether the same failure-trace signals are informative on other providers, benchmarks, or budget levels is an open empirical question. The most efficient next replication would be a second GSM8K provider run or a same-provider second benchmark with the identical frontier, L1, S1, and TALE logging contract. Section 6.1 enumerates the scope boundaries explicitly.

6.1 Limitations

We summarize what is established by the current evidence and then state the main scope boundaries explicitly.

Established in this submission. The policy is runtime-legal and gold-free by construction (Table 15); evaluation is disjoint by design across the Aggregate-720 sources; and FTA leads each individual external baseline (L1/S1/TALE) by point estimate with strictly positive source-stratified CI lower bounds on Aggregate-720. The pooled four-answer comparison is also reported and shows a closer gap whose confidence interval includes zero on both evaluation sets.

Benchmark scope. Results are scoped to GSM8K (`openai/gsm8k`) under the matched-budget protocol described in Section 4. No accuracy claim is made beyond this dataset and task family.

Provider-specific scope. All main evidence is obtained on Cohere command-r-plus-08-2024. The method is not claimed to generalize to other providers, model families, or API versions without independent replication.

Seed 61 is failure-enriched. The 120-example seed 61 split was used as a development seed and is failure-enriched: all four methods (frontier, L1, S1, TALE) fall to approximately 54–58% accuracy on this split, substantially below the 81–87% range on seeds 41 and 71. Absolute accuracy estimates from seed 61 alone are therefore not representative of normal evaluation conditions. Seed 61 is included in Aggregate-720 for aggregate coverage, and the source-stratified confidence intervals account for its higher difficulty, but it should not be interpreted as an independent IID validation split.

Pooled-ensemble comparison not statistically separated. Against a pooled four-answer majority-vote ensemble (Pooled-4), FTA leads by point estimate on both evaluation sets (+2.33 pp on Final-300; +0.83 pp on Aggregate-720), but the paired bootstrap confidence interval includes zero in both cases: $[-0.67, +5.67]$ on Final-300 and $[-1.11, +2.78]$ on Aggregate-720. FTA cannot be claimed to be statistically superior to the pooled ensemble. This comparison is reported as a robustness check, not as a retained superiority claim.

Budget scope and compute accounting. The matched-budget claim is a per-method evaluation contract. Full candidate-pool construction requires $4 \times B = 24$ logical inference calls per example; FTA itself adds no post-generation model calls. Monetary cost depends on provider pricing and is outside the main comparison metric.

D9 multi-provider evidence is supporting only. The preliminary D9 experiment provides supporting evidence that the failure-trace signal may generalize across providers, but the evaluation is restricted to 80–160 examples per provider and uses cross-validation rather than a held-out test set at full scale. D9 findings are Tier 2 supporting evidence; they are not a second primary submission claim.

Tie-breaker sensitivity. The TALE$\succ$S1$\succ$L1 fallback used when all three external answers differ is deterministic and gold-free, but it is not claimed to be optimal. Alternative tie orders move Aggregate-720 replay by less than one percentage point in the available artifacts.

Residual pool-miss failures. A residual failure class exists in which all methods—frontier and external baselines—produce incorrect answers for the same example. The proposed policy cannot recover from this class, as no correct answer is present in the candidate pool.

Not-retained variants. Follow-up prototypes were explored during development but did not meet validation-grade evidence criteria. They are reported as negative results in Table 10 and are not part of the retained method.

7 Conclusion

We presented the **Failure-Trace Allocator (FTA)**, a deterministic, gold-free answer-selection policy for matched-budget LLM inference. On the primary Aggregate-720 evaluation (three disjoint seeds, zero overlap), FTA achieves 80.69% (581/720) versus L1 77.64%, S1 77.08%, and TALE 75.14%. Source-stratified 95% bootstrap confidence intervals are strictly positive for all four external baseline comparisons, with the lower bound versus the best external at +1.11 pp. On the Final-300 evaluation (seed 71, $n = 300$), FTA reaches 86.67% (260/300), compared to 83.00% for L1, 82.00% for S1, and 78.33% for TALE.

The policy consists of two rule-based gates applied in strict priority order. The Low-Depth Guard (LDG) detects structurally weak frontier search branches using logged failure-trace metadata and substitutes the external majority answer. The External-Consensus Fallback (ECO), a low-frequency high-precision safeguard in this evaluation, identifies cases where all three external baselines unanimously disagree with the frontier's selection, overriding to the consensus answer. Both gates are gold-free: no correctness labels, example identifiers, or artifact paths are accessed at runtime, and no additional inference calls are issued after candidate generation. The rules, trigger conditions, and exact field names are fully specified in Table 6 and Table 1, with reproducibility details documented in the appendix.

As supporting preliminary evidence, a multi-provider regime-learning experiment (D9) using the same failure-trace signal achieves a cross-validated accuracy of 0.5018 ± 0.0252 against a frontier baseline of 0.3436, with zero false-override events across all tested providers and thresholds. These preliminary results are consistent with the failure-trace signal having provider-independent utility, though a full multi-provider evaluation at matched scale remains for future work.

Failure analysis of the 40 residual errors on Final-300 shows that 24 (60%) are *all-methods-wrong* cases in which no evaluated method produced the correct answer; the dominant subtype is multi-step composition failures (10/24) where the error is embedded in intermediate reasoning rather than in the final selection step. Further accuracy gains in this setting are therefore likely to require advances in candidate generation rather than selection policy alone.

Future directions include learned allocation policies that replace the deterministic gates with calibrated classifiers, evaluation on broader datasets and provider families, stronger candidate-pool construction strategies that reduce the pool-miss failure class, and extension to multi-turn or open-ended generation settings where failure-trace signals may take different structural forms.

Acknowledgements

The author thanks Professor Ioannis Koutis for his support, guidance, and helpful discussions during this work. The author is also deeply grateful to his mother for her continuous support and encouragement. The author acknowledges Cohere and CloudRift AI for providing research API credits that supported the model-inference experiments in this study. The author also thanks Anders Borum for providing free lifetime access to Secure ShellFish, which supported the remote research workflow.

Table 10: Negative results and not-retained outcomes.

Prototype	Result	Implication
FIX-5 TALE-default router	Failed as preferred policy on larger unbiased validation; no reliable transfer to retained setting.	Keep Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4) as main method; do not route by TALE-default in deployment claim.
FIX-6 extra-action variant	Independent follow-up run showed negative net vs Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4) (extra_frontier −1, extra_tale −4).	No accuracy-changing extra-action policy retained.
FIX-7 cluster selector	Weak final300 gain (+0.67 pp) with not-retained decision.	Prototype only; insufficient for retention in the main method.
FIX-8 robust parser/canonicalizer	0 recoveries and 0 aggregate gain (final300 and aggregate720 deltas both 0).	No parser-driven policy change justified.
FIX-9 pool-miss detector audit	23/24 cases generation-likely; selector-only opportunity limited.	Residual error frontier is mostly candidate-generation limited under current logs.

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A Reproducibility and Supplementary Material

This appendix collects reproducibility information, additional result decompositions, parameter grounding, runtime feature legality, and paper-facing name mappings.

Table 11: Reproducibility checklist.

Item	Status	Value	Artifact/reference
Dataset	complete	`openai/gsm8k` primary subsets (300+120+300)	manifests + `aggregate_with_prior_validation.csv`
Provider / model	complete	Cohere / command-r-plus-08-2024	`validation_manifest.json`
Seeds	complete	41, 61, 71 (primary); 31 sensitivity	`aggregate_with_prior_validation.csv`
Budget	complete	$B = 6$ matched across all methods	manifests + per-example rows
Output roots	complete	major roots enumerated	supplementary reproducibility artifact + `docs/LATEST_RESULTS_AND_CLAIMS.md`
Disjointness proofs	complete	source overlaps = 0	`final_postrun_metrics.json` checks
Dry-run logs	complete	dry-run plans present for all validation runs	`dry_run_call_plan.log`
Postrun outputs	complete	final postrun metrics and CI tables present	`final_fix24_all_external_postrun` root
Bootstrap method	complete	5000 resamples; pooled + source-stratified summaries	`bootstrap_ci_summary.csv`
Policy code files	complete	`support_aware_selector.py`, `cluster_answer_selector.py`, `robust_answer_parser.py`	`experiments/`
Tests	partial	policy/parser tests present; not all scripts formalized as tests	`tests/`
Ablation variants	complete	FIX-5/6/7/8/9 documented with negative or insufficient evidence	Table 10 + supplementary documentation
Rerun requirements	partial	Provider API access required for live reruns	validation scripts/manifests

Additional Result Decompositions

The following tables give the per-example win/loss/tie decomposition (Table 12) and supported versus unsupported claim examples (Table 13).

Table 12: Win/loss/tie decomposition for Failure-Trace Allocator (FTA) versus each external baseline on Final-300 (seed 71, budget 6, $n = 300$). A win is a case where Failure-Trace Allocator (FTA) is correct and the baseline is not; a loss is the reverse; a tie is when both policies agree on correctness. The low loss counts confirm that Failure-Trace Allocator (FTA) recoveries substantially outweigh regressions, particularly vs. TALE.

Comparison	Wins	Losses	Ties
Failure-Trace Allocator (FTA) vs L1	19	8	273
Failure-Trace Allocator (FTA) vs S1	23	9	268
Failure-Trace Allocator (FTA) vs TALE	27	2	271

Table 13: Supported and unsupported claim examples. "Safe" claims are within the evidence scope; "Unsafe" claims are not.

Type	Claim statement
Safe	Failure-Trace Allocator (FTA) outperforms L1/S1/TALE and best-external on aggregate720 (matched budget), with positive source-stratified bootstrap CI lower bounds for all baseline comparisons.
Safe	Failure-Trace Allocator (FTA) is ahead of pooled-answer ensemble baselines by point estimate on Final-300 and Aggregate-720, but the confidence intervals include zero.
Safe	Failure-trace-guided gold-free rules recover low-depth and external-consensus failure modes.
Safe	FIX-5/6/7/8/9 are not retained due to insufficient or negative evidence.
Unsafe	Universal superiority over all reasoning methods.
Unsafe	Statistically separated superiority over pooled-answer ensemble baselines.
Unsafe	Universal gains across all math datasets/providers.
Unsafe	Promoted gains from FIX-7/FIX-8/FIX-9.
Unsafe	Dollar/latency savings without direct measurement.

Parameter Grounding and Sensitivity

Table 14 documents the key parameters and design choices used in Failure-Trace Allocator (FTA), including their values, roles, and available robustness evidence.

Table 14: Parameter grounding for Failure-Trace Allocator (FTA) and the evaluation protocol. We report parameter grounding and available robustness checks rather than a full continuous hyperparameter sweep; a full threshold sweep is left for future work. All values are fixed and applied identically across all evaluated splits.

Parameter / Design choice *(value used)*	Role, justification	Robustness evidence available	Remaining limitation
Budget ($B = 6$)	Hard cap on logical API calls per example, ensuring matched-budget comparison. Fixed by provider run protocol; all methods use the same B.	Sensitivity source (+100 examples) included as separate labeled set; B held constant.	No multi-B sweep reported.
Low-depth guard: `single_weak _frontier_branch` *(string match; implementation constant)*	Identifies frontier answers from structurally weak search paths. Derived from failure-trace analysis of frontier run logs (`support_aware_selector.py`).	Trigger frequency and recovery/regression counts available in postrun artifacts.	No threshold variation sweep in manuscript.
Low-depth guard: `frontier_support`$= 0 \wedge$ `pool_group_count`≥ 2 *(integer thresholds: 0 and 2)*	Detects zero-support frontier outputs when the pool contains ≥ 2 distinct answer groups. Conservative: requires complete absence of support plus pool diversity.	Condition derived from offline evaluation review; trigger counts in supplementary artifacts.	No continuous threshold sweep.
External-consensus condition (ECO): `l1`=`s1`=`tale` $\wedge$ all $\neq$ frontier *(unanimity gate)*	Identifies unanimous external agreement diverging from frontier. Unanimity is a strong, conservative signal; partial agreement is not acted on.	Trigger frequency available from postrun replay artifacts.	No partial-consensus variant evaluated.
Bootstrap resamples *($n = 5{,}000$)*	Controls Monte Carlo variance in paired bootstrap CI estimation. Standard practice for stable 95% CIs at $n \leq 720$.	Single-seed results show consistent sign agreement.	No resample-count sensitivity sweep reported.
Evaluation seeds / splits *(seeds 41, 61, 71)*	Define disjoint aggregate and source-stratified analysis. Seeds span independent run batches; disjointness verified by example ID and question hash.	Source-stratified CIs show consistent positive sign for all baselines.	Single task/provider; no cross-dataset seed check.
Disjointness check *(overlap = 0)*	Prevents data leakage between primary sources. Verified programmatically on all three source sets.	Confirmed in postrun evaluation artifacts; overlap count = 0.	Within GSM8K only; no cross-dataset overlap check.
Provider / model *(Cohere command-r-plus-08-2024)*	Inference backend for all runs. Chosen for API availability and run stability during the experiment period.	All compared methods use the same provider/model.	Provider-specific; generalization not established.
External tie fallback *(TALE$\succ$S1$\succ$L1)*	Deterministic gold-free fallback when L1, S1, and TALE all disagree. Retained as the fixed retained-policy convention rather than optimized on Aggregate720.	Alternative LDG-only replays on Aggregate720 give 80.14% for S1-first and 80.42% for L1-first; frontier fallback on all-different external ties gives 80.83%.	Tie order is not separately validated as an optimal policy.

Runtime Feature Legality

Table 15 lists all features used by Failure-Trace Allocator (FTA) at inference time and confirms their gold-free status.

Table 15: Runtime feature legality for Failure-Trace Allocator (FTA). All gating inputs are logged as side-effects of normal frontier execution and are available at inference time without any access to gold answers, correctness labels, example IDs, or artifact paths.

Feature	Source	Gold-free?	Why it is allowed
`override_reason`	Frontier run metadata	Yes	Logged search-state signal; e.g., `"single_weak_frontier_branch"` or `"direct_frontier_agree"`; not a correctness label.
`frontier_support`	Frontier run metadata	Yes	Integer support count from the frontier search; derived from pool structure, not correctness.
`candidate_pool_answer_group_count`	Frontier run metadata	Yes	Number of distinct answer groups in the pool; reflects diversity, not correctness.
`frontier_answer`	Frontier generator output	Yes	Frontier candidate selected before gating; proposed output only.
`l1_answer`	External L1 baseline output	Yes	L1 inference output, not the ground-truth label.
`s1_answer`	External S1 baseline output	Yes	S1 budget-forcing output only.
`tale_answer`	External TALE baseline output	Yes	TALE prompt-budgeting output only.
Excluded at runtime	N/A	N/A	`gold_answer`, `exact_match`, `example_id`, and artifact paths are never used in routing.

Paper-Facing Name Mapping

Table 16 maps paper-facing short names to their implementation identifiers, roles, runtime availability, and gold-free status.

Table 16: Paper-facing name mapping. Short names used throughout the manuscript and their implementation identifiers, roles, runtime availability, and gold-free status. Raw identifiers are retained in Table 1, Table 6, and Table 15.

Paper-facing name	Implementation identifier	Role	Used at runtime?	Gold-free?
failure-trace-guided allocator	`Failure-Trace Allocator (FTA, implementation: FIX-2+FIX-4)`	Main two-gate policy	Yes	Yes
weak-frontier trigger	`single_weak_frontier_branch`	Low-Depth Guard condition; `override_reason` value	Yes	Yes
direct-agreement trigger	`direct_frontier_agree`	External-Consensus Fallback condition; `override_reason` value	Yes	Yes
L1	`external_l1_max`	External baseline; L1-max answer selector	Yes	Yes
S1	`external_s1_budget_forcing`	External baseline; S1 budget-forcing method	Yes	Yes
TALE	`external_tale_prompt_budgeting`	External baseline; TALE prompt-budgeting method	Yes	Yes
best external	best external comparator	Highest-accuracy external baseline on a given split	No	Yes

Data Availability

The GSM8K benchmark questions used for evaluation are available from the original source [1]. A supplementary reproducibility package is included with this submission and contains the processed evaluation artifacts used to generate the reported tables and figures (including aggregate summaries, paired comparison tables, and run manifests). The reported Aggregate-720 accuracy (80.69%) and associated confidence-interval tables can be reproduced from these processed artifacts without issuing new provider API calls. No API keys, credentials, or other secrets are included in the supplementary package.

Code Availability

The supplementary package includes the code paths and scripts used for policy replay and evaluation of the reported artifacts. Live regeneration of candidate answers requires provider API access to Cohere command-r-plus-08-2024; however, the manuscript's reported post-generation analyses are reproducible offline from the supplied processed artifacts.

Funding

This work was supported by research API credits from Cohere and CloudRift AI. No other specific grant funding was received from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests

The authors declare no competing interests.

Use of AI Tools

The author used ChatGPT, Cursor, Claude, GitHub Copilot, Cohere models, and Perplexity AI during parts of the coding, literature-search, manuscript-drafting, editing, and artifact-preparation workflow. These tools were used for assistance with code navigation, implementation support, language editing, reference discovery, and drafting support. All scientific claims, experimental design choices, code changes, analyses, reported results, citations, and final manuscript text were reviewed and approved by the author, who takes full responsibility for the content of the paper.

Author Contributions

Soroush Vahidi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review and editing, Visualization.